PROJECT PROPOSAL

Temporary Rental Space Management System of MFU

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IN COMPUTER ENGINEERING

MAE FAH LUANG UNIVERSITY

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A COMPUTER ENGINEERING PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE DEGREE OF
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TO BE PARTIAL FULFILLMENT OF REQUIREMENTS FOR
THE DEGREE OF BACHELOR OF ENGINEERING IN
COMPUTER ENGINEERING

2026

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Abstract

This senior project proposes the development of a web-based Temporary Rental Space Management System for Mae Fah Luang University. Designed to serve both internal university members and external organizations, the system addresses the inefficiencies of traditional, paper-based reservation methods. By fully digitizing the workflow, the platform aims to eliminate manual processing errors, reduce communication delays, and prevent scheduling conflicts.

The system provides a comprehensive online solution where users can check real-time availability, select customized add-ons, and process secure payments seamlessly. Simultaneously, administrators are equipped with a centralized dashboard for dynamic facility management, digital approval workflows, and data visualization. Ultimately, this system enhances the transparency, convenience, and overall operational efficiency of managing the university's commercial and non-commercial rental spaces.

Keyword: Space Management System, Web Application, Online Booking, Payment Integration, Mae Fah Luang University

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CHAPTER 1
INTRODUCTION

1.1 Background and Rationale

The Property Management Division of Mae Fah Luang University is primarily responsible for managing and monetizing a variety of temporary rental spaces. These facilities serve as a crucial source of revenue for the university, being utilized not only for internal academic activities but also for commercial rentals by external individuals, companies, and organizations.

However, the current space reservation process relies heavily on manual, paper-based procedures. This traditional approach is highly susceptible to human error, frequently resulting in scheduling conflicts, delayed communications, and significant complexities in managing financial transactions and official documentation with external clients.

To resolve these operational bottlenecks and enhance user convenience, this project proposes the development of a comprehensive web-based Temporary Rental Space Management System. By providing a fully digital platform that allows external users to easily book spaces and process payments online, the system will significantly reduce human errors and redundant manual tasks. Ultimately, this solution will empower the Property Management Division to maximize resource utilization and manage revenue generation with high accuracy and efficiency.

1.2 Objectives

1.2.1 To develop a fully functional web-based temporary rental space management system for Mae Fah Luang University.
1.2.2 To implement a full-stack system architecture, demonstrating the seamless integration between frontend development and backend management.
1.2.3 To enable users to process secure online bookings and provide administrators with a centralized dashboard for efficient facility management.

1.3 Scope

1.3.1 System Users

Users are categorized into distinct authorization levels based on their relationship with Mae Fah Luang University (MFU). The system will automatically apply different rental pricing tiers based on these roles:

- 1.3.1.1 Internal Users: Individuals or entities operating within the MFU organization. They will be charged the "Internal Rate".
- 1.3.1.2 Collaborative Users (Co-op): Users involved in cross-organizational partnerships or activities co-hosted with the university. They will be charged the "Collaborative Rate".
- 1.3.1.3 External Users: Individuals or private companies outside the MFU organization. They will be charged the standard "External Rate".
- 1.3.1.4 System Administrator: Personnel responsible for overseeing facility data, managing reservation requests, and handling administrative workflows.

1.3.2 User Functional Scope:

- Availability Search & Advanced Booking: Check real-time room availability and submit advance reservation requests within specified timeframes.
- Service Customization (Add-ons): Select supplementary services or additional equipment (e.g., specific quantities of tables and chairs) during the booking process, with automated calculation of additional costs.
- Secure Payment Integration: Process transactions securely and seamlessly via an integrated payment gateway.
- Promotions and Incentives: Apply promotional discounts or special offers provided by the system during the reservation process.
- Reservation Management & History: Access personal booking logs, specify reservation purposes, track current statuses, and cancel existing bookings according to system policies.
- Status and Event Notifications: Receive automated alerts regarding booking confirmations, workflow status updates, and announcements for new promotional campaigns.
- Post-Service Feedback: Submit reviews and satisfaction ratings after completing the reservation lifecycle to provide service evaluations.
- **Secure Authentication:** Users can securely log in via Google OAuth. The system automatically assigns user roles (e.g., internal or external) based on their registered email domains, such as @mfu.ac.th or @lamduan.mfu.ac.th.
- **Secure Payment Integration:** Users can process secure and instant online payments via a third-party gateway, specifically utilizing dynamic PromptPay QR Codes.

1.3.3 Administrator Functional Scope:

- Facility & Rate Management: Maintain room records (Add/Edit/Disable) and manage pricing structures. This includes the ability to bulk import room details and room rates directly from spreadsheet datasets (Excel files).
- Administrative Dashboard & Data Visualization: Access a centralized analytics dashboard featuring comprehensive data visualization through interactive graphs and charts. This tool allows for monitoring real-time booking statistics, revenue tracking, and facility utilization, with the ability to drill down into specific data entries for detailed analysis.
- Approval Workflow and Documentation: Process reservation requests by updating statuses (Approve/Disapprove). This includes the ability to import executive authorization files and export official approval documents to serve as verified digital evidence.
- Promotion Management: Create, configure, and manage promotional campaigns to encourage system utilization.
- System Notifications & Broadcasts: Manage automated system alerts and broadcast new promotional information or critical updates to designated user groups.

1.4 Methodology

This project follows an iterative Software Development Life Cycle (SDLC) to ensure the system meets the requirements of Mae Fah Luang University.

- 1.4.1 Requirement Gathering and Analysis: Collect and analyze user requirements by reviewing existing manual reservation processes and examining documents from the Intellectual Property and Innovation Management Division.
- 1.4.2 System Design: Design the system architecture, user interface (UI), and database. This includes designing the relational database schema and mapping data relationships using PostgreSQL and pgAdmin4.
    - 1.4.2.1 Third-Party Integrations (Overview): The system uses Google OAuth for secure Single Sign-On (SSO) and integrates a commercial Payment Gateway (e.g., Opn Payments or 2C2P) to process dynamic PromptPay QR Codes. Utilizing this payment service incurs a standard transaction fee of approximately 1.65%.
- 1.4.4 System Testing and Evaluation: Conduct rigorous testing to identify and resolve bugs. This includes testing role-based access for internal and external users, validating the approval workflows, and ensuring the system prevents scheduling conflicts.
- 1.4.5 Documentation and Presentation: Prepare the formal Senior Project 1 and 2 documentation and defend the project before the examination committee.

1.5 Plan

- **Table 1.1** Working plan

	Senior project 1					Senior project 2				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Proposal preparation and defense										
Literature review and requirement gathering										
System analysis										
System design										
Senior project 1 document preparation										
Senior Project 1 examination										
System development										
System testing and evaluation										
Senior project 2 document preparation										
Senior Project 2 examination										

1.6 Expected Result

1.6.1 A fully functional and secure web-based room reservation system for Mae Fah Luang University.
1.6.2 A streamlined booking experience with integrated online payment, enhancing convenience for external commercial users.
1.6.3 Significantly reduced human errors and redundant paperwork in the reservation and financial tracking processes.
1.6.4 Improved operational efficiency and enhanced revenue generation capabilities for the Property Management Division.

1.7 Place and Equipment

1.7.1 Equipment

1.7.1.1 Hardware

- Personal computer or laptop
- Internet connection

1.7.1.2 Software

- Frontend: Vue.js, Tailwind CSS
- Backend: Node.js
- Database: PostgreSQL, pgAdmin4

1.7.1.3 Data set

- University room details and specifications

CHAPTER 2
LITERATURE REVIEW

This chapter reviews the existing literature, systems, and technologies relevant to the development of the Temporary Rental Space Management System. It evaluates the current booking methods at Mae Fah Luang University (MFU) and explores the core technologies and third-party integrations utilized in this project.

2.1 Review of the Existing MFU Room Booking System

Currently, the room booking process at Mae Fah Luang University relies significantly on manual workflows and basic internal platforms. These existing systems are primarily designed for internal academic use, lacking the flexibility to accommodate external commercial clients. Furthermore, the absence of an integrated online payment gateway forces users to complete transactions manually, leading to administrative delays.

To highlight the improvements proposed by this project, a comparison between the existing MFU system and the proposed system is presented in Table 2.1.

Table 2.1 Comparison between the Existing MFU Booking System

Features	Existing MFU Booking System	Proposed System
External User Booking	Manual paper-based application and in-person coordination	Direct online booking via web platform
Overall Workflow	Paper-based routing with manual administrative approval	Fully automated web-based approval workflow
Authentication	Local university database login (Staff/Students only)	Google OAuth (SSO) for all users
Payment Method	Manual transactions / Cash at the counter	Integrated Payment Gateway (PromptPay QR)
Service Add-ons	Manual request required during paperwork submission	Automated add-on selection during booking

2.2 Related Technologies and Integrations

To achieve the proposed features, several modern technologies and third-party services were reviewed and implemented:

- **2.2.1 Web Application Frameworks:** Vue.js and Tailwind CSS were selected for the frontend to provide a responsive and dynamic user interface. Node.js and PostgreSQL were chosen for the backend to ensure robust data management and high-performance server processing.
- **2.2.2 Single Sign-On (SSO) and Authentication:** Google OAuth is utilized to handle user authentication securely. This eliminates the need for localized password storage, leveraging Google's robust security infrastructure while providing a seamless login experience for users with existing university or personal Google accounts.
- **2.2.3 Payment Gateway Providers:** For the financial transaction module, Opn Payments (formerly Omise) was selected over other regional providers like 2C2P. Opn Payments offers a highly developer-friendly API, seamless integration with Node.js, and a highly competitive transaction fee for dynamic PromptPay QR Codes (1.65%). This makes it the most cost-effective and efficient solution for the university's Property Management Division.

2.3 Comparison with Commercial Booking Platforms

To ensure the proposed system meets standard practices while addressing the university's specific operational needs, commercial platforms such as Peerspace and Skedda were reviewed. While these platforms offer robust general features, they lack customized workflows for internal university approvals and role-based pricing structures.

Table 2.2 Comparison between the Proposed System and Commercial Platforms

Feature	Proposed MFU System	Peerspace	Skedda
Primary Focus	University properties and facilities	General event spaces and studios	Academic & office workspaces
Approval Workflow	Custom Admin Approval (Management Office)	Host Approval	Instant Booking / Admin override
Pricing Tier	Dynamic (Internal / External / Co-op)	Standard Public Rate	Standard Public Rate
Payment Gateway	PromptPay QR (Opn Payments)	Credit Card (Stripe)	Online Card Payments
Service Add-ons	Tailored to MFU (e.g., specific tables, mics)	General host offerings	Limited custom add-ons

CHAPTER 3
ANALYSIS AND DESIGN

3.1 Use case diagram

The use case could be demonstrated by a use case diagram in **Figure 3.1**

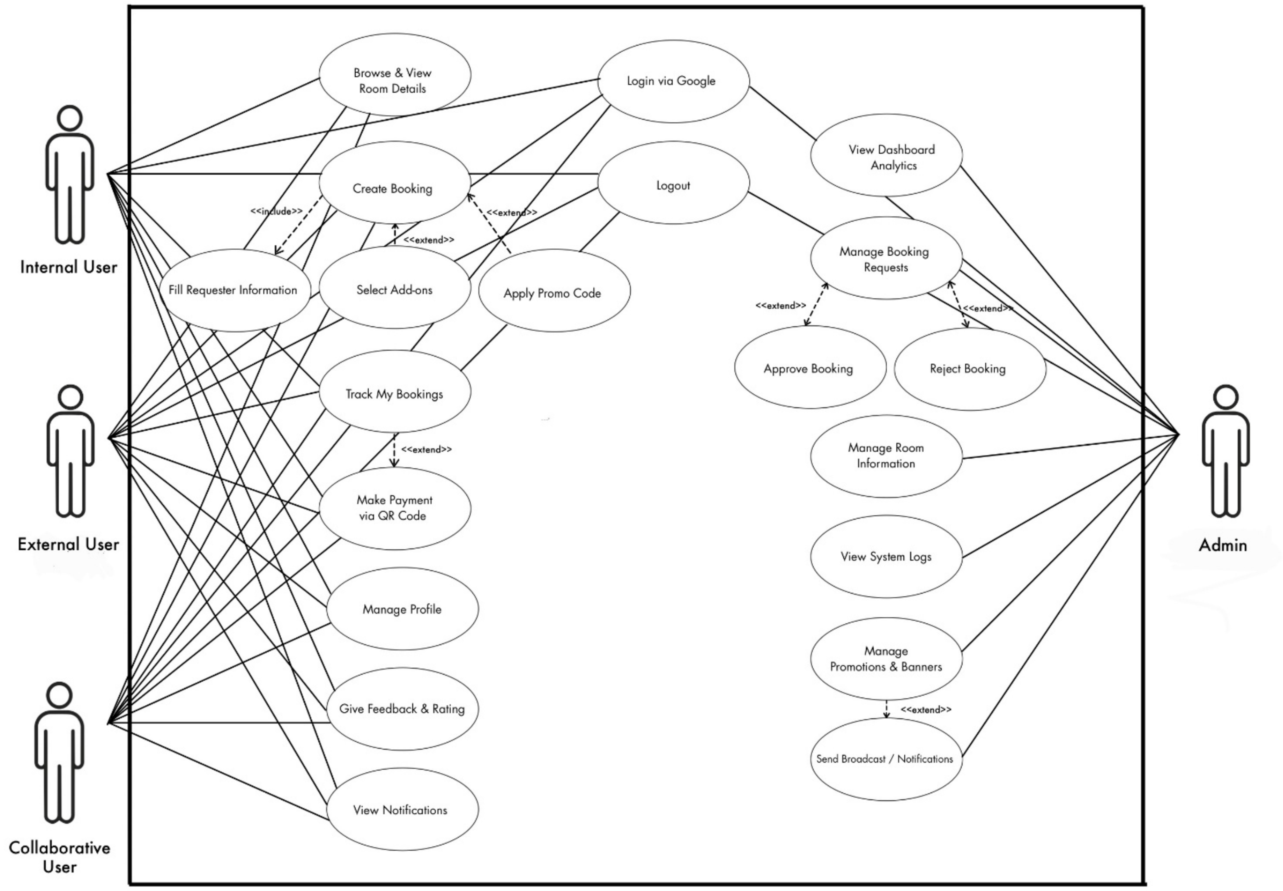

Figure 3.1 Use case diagram for existing website

3.2 Swim lane diagram

The login could be demonstrated by a swim lane diagram in **Figure 3.2**

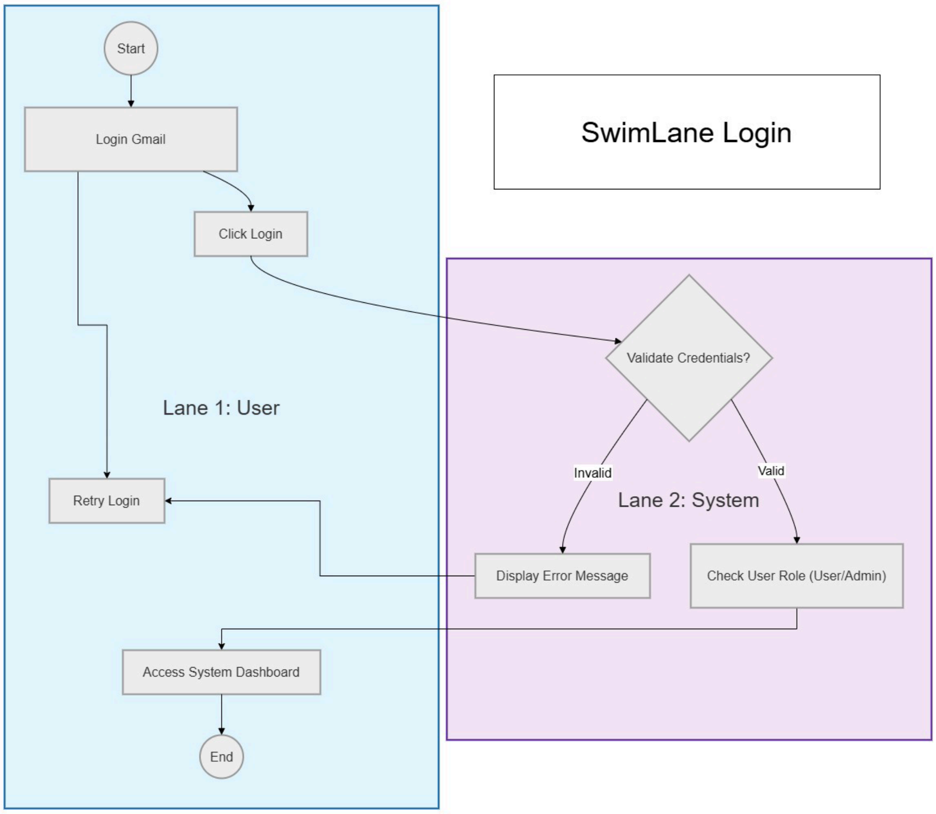

Figure 3.2 Swim lane diagram for login

The booking could be demonstrated by a swim lane diagram in **Figure 3.3**

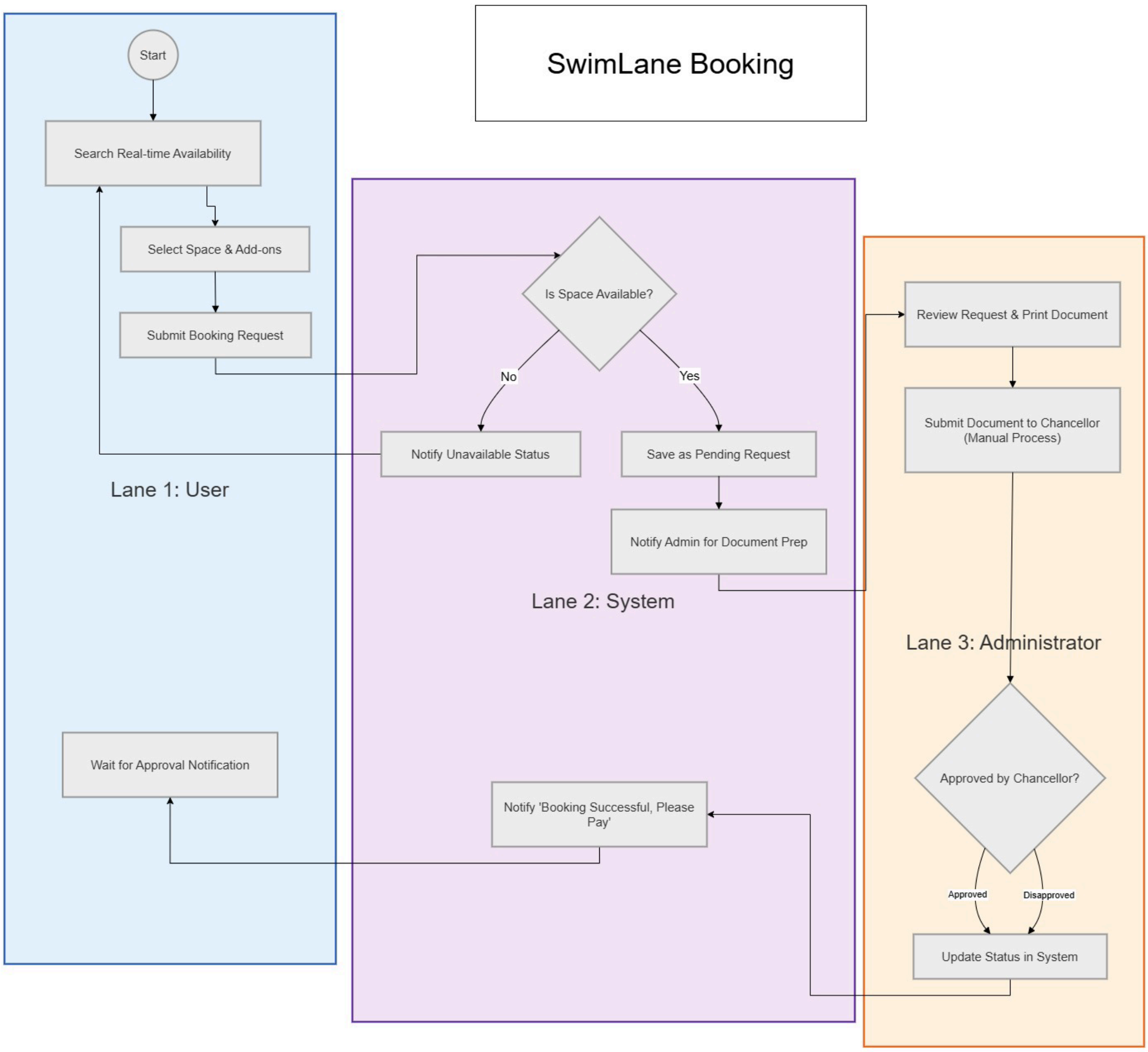

Figure 3.3 Swim lane diagram for booking

The payment could be demonstrated by a swim lane diagram in **Figure 3.4**

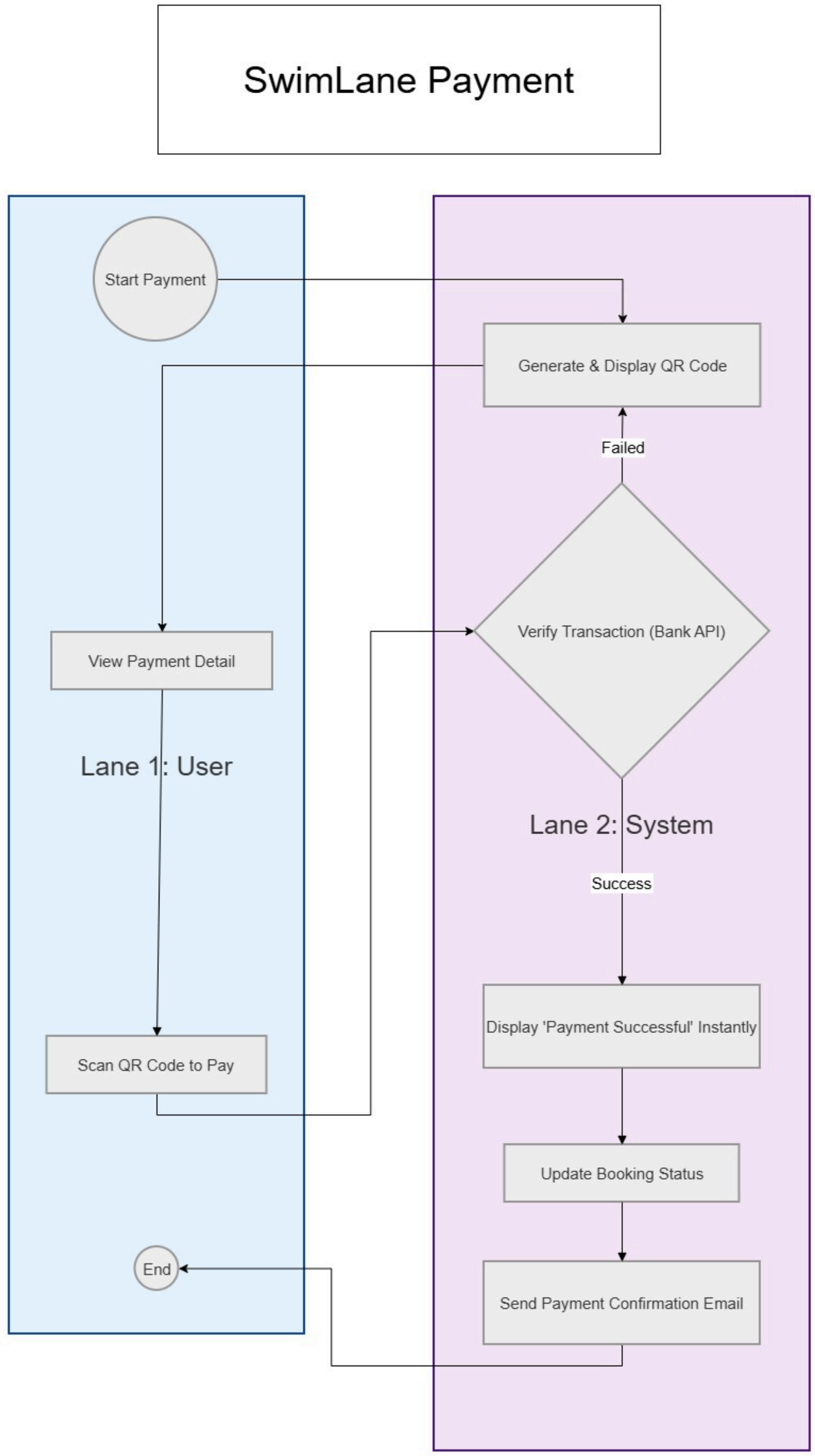

Figure 3.4 Swim lane diagram for payment

3.3 Site map

The webpage could be demonstrated by a site map in **Figure 3.5**

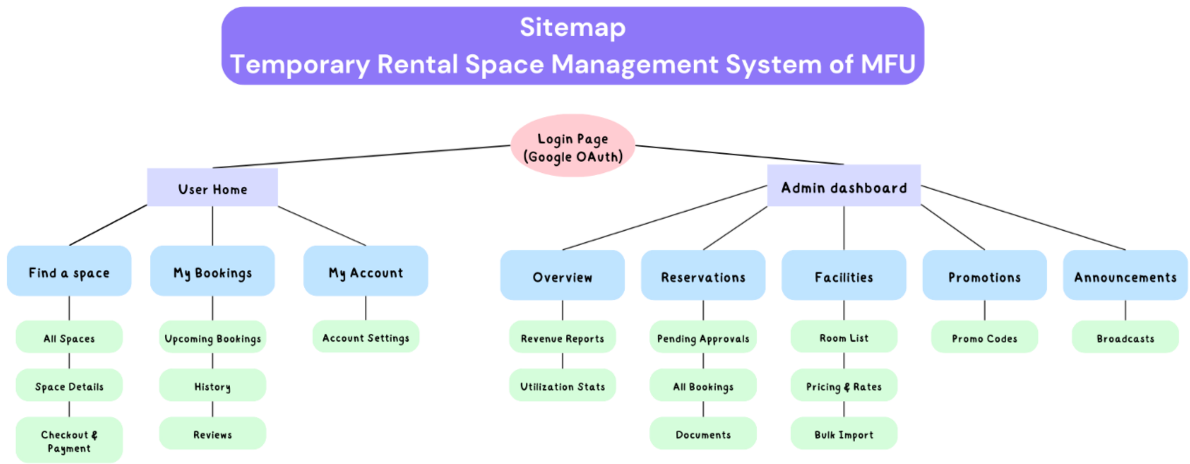

Figure 3.5 Site map for webpages

3.4 User Interface (UI)

The login page could be demonstrated by a user interface in **Figure 3.6**

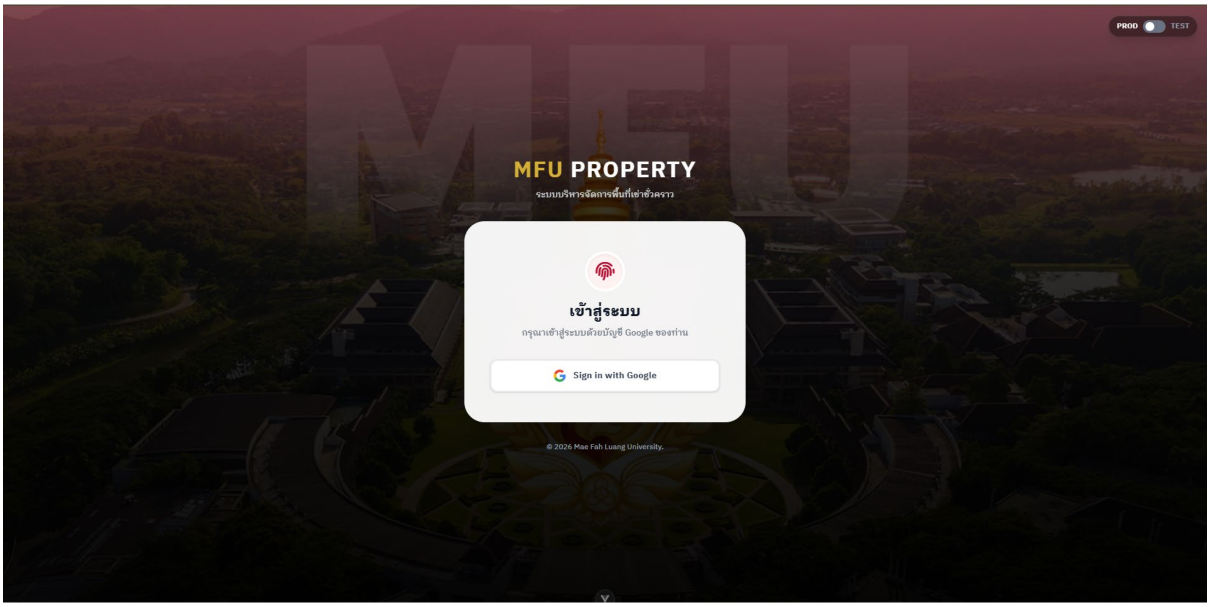

Figure 3.6 User interface for login page

The home page could be demonstrated by a user interface in **Figure 3.7 - Figure 3.10**

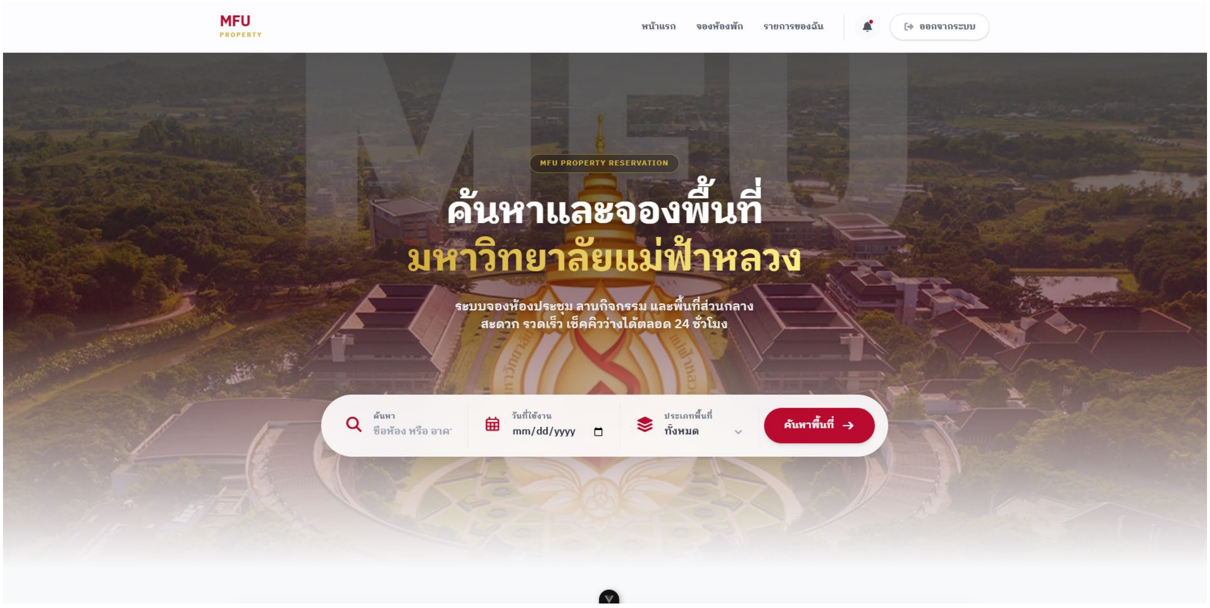

Figure 3.7 User interface for user home page

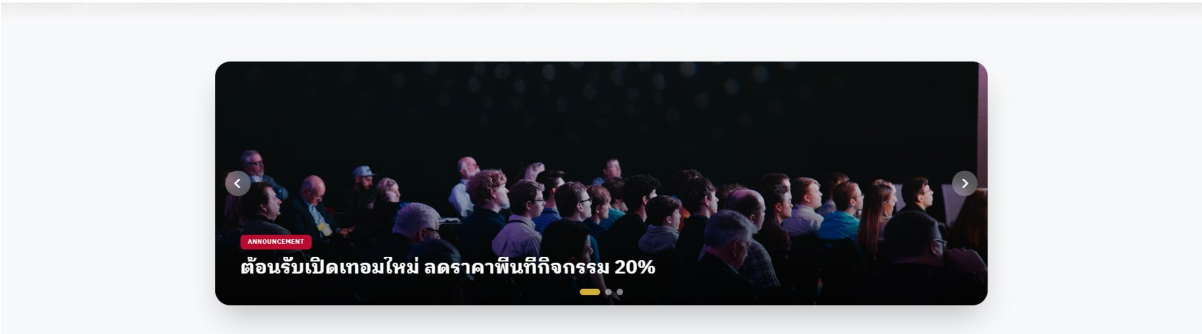

Figure 3.8 User interface for user home page

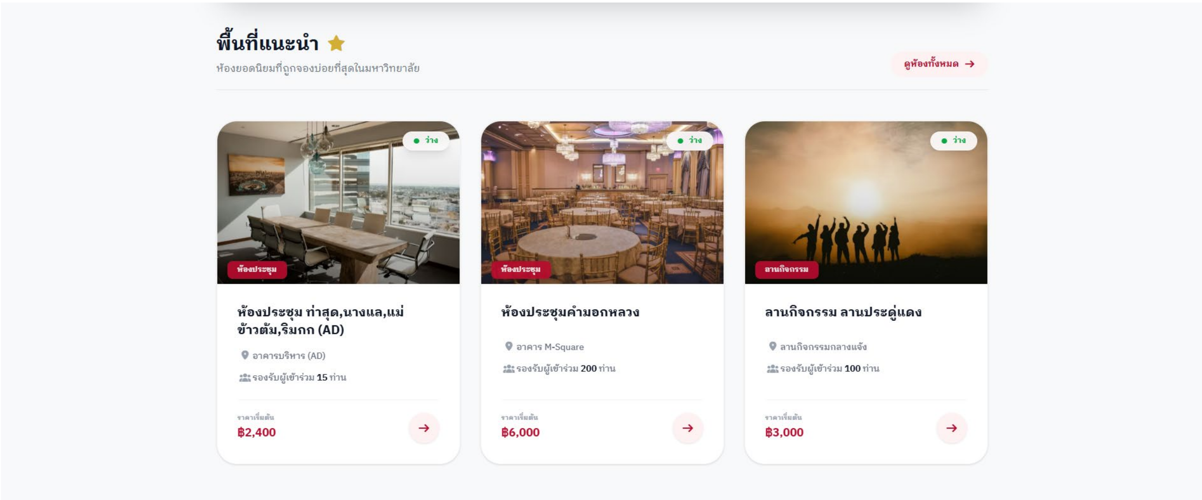

Figure 3.9 User interface for user home page

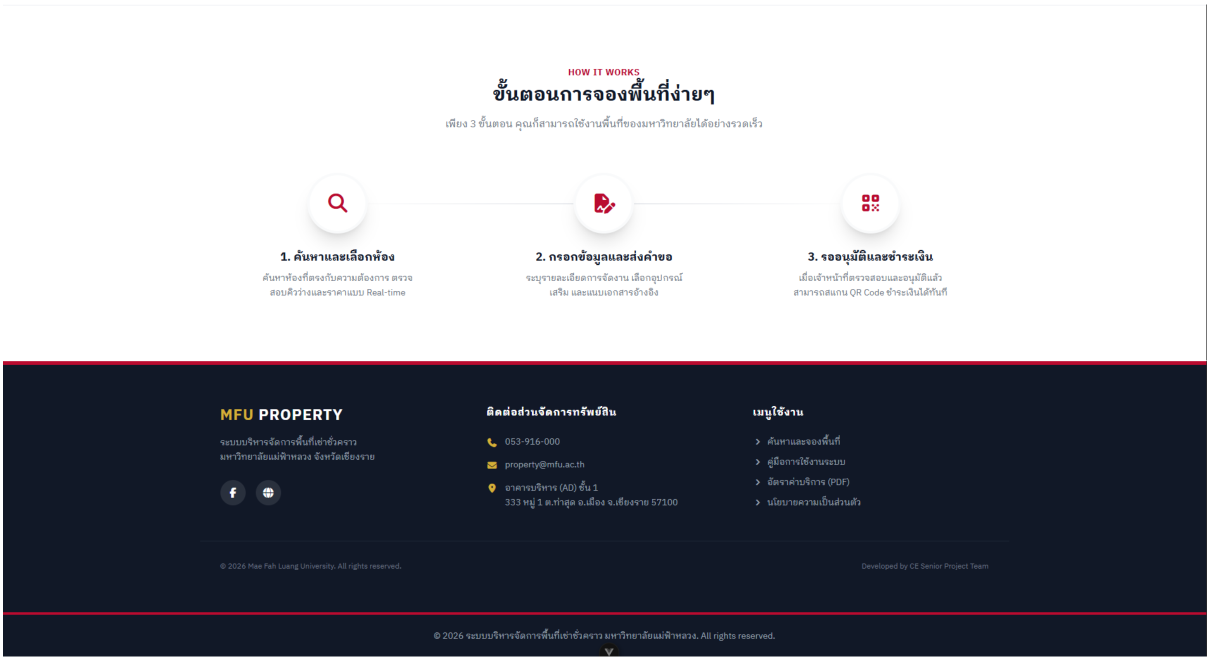

Figure 3.10 User interface for user home page

The all-room page could be demonstrated by a user interface in **Figure 3.11 - Figure 3.13**

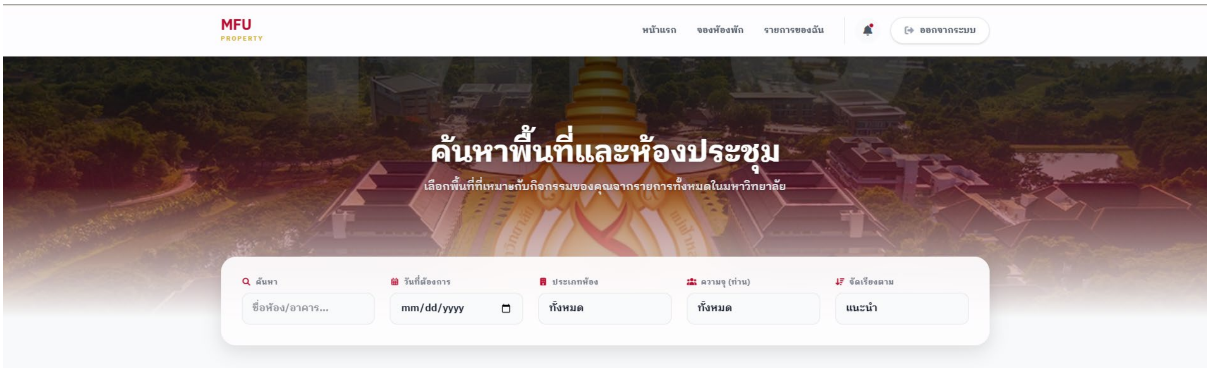

Figure 3.11 User interface for all-room page

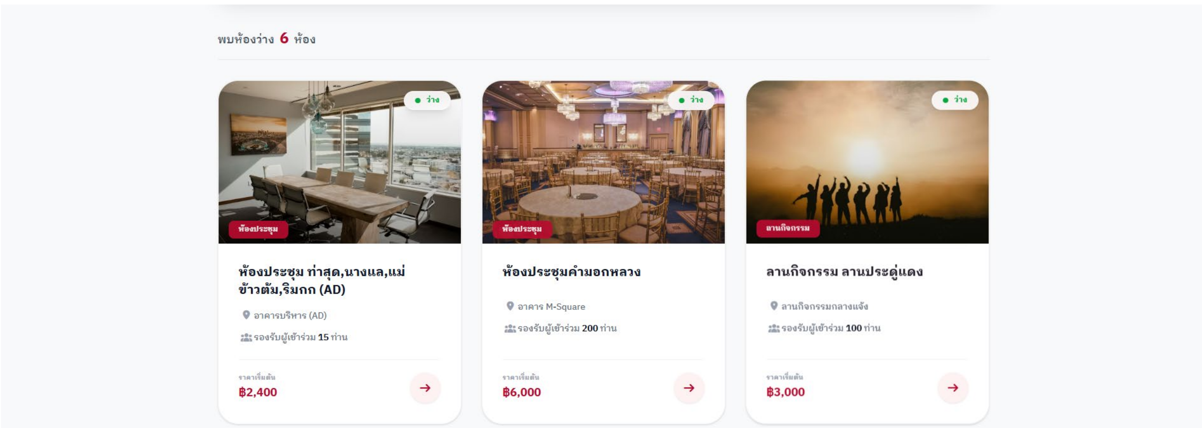

Figure 3.12 User interface for all-room page

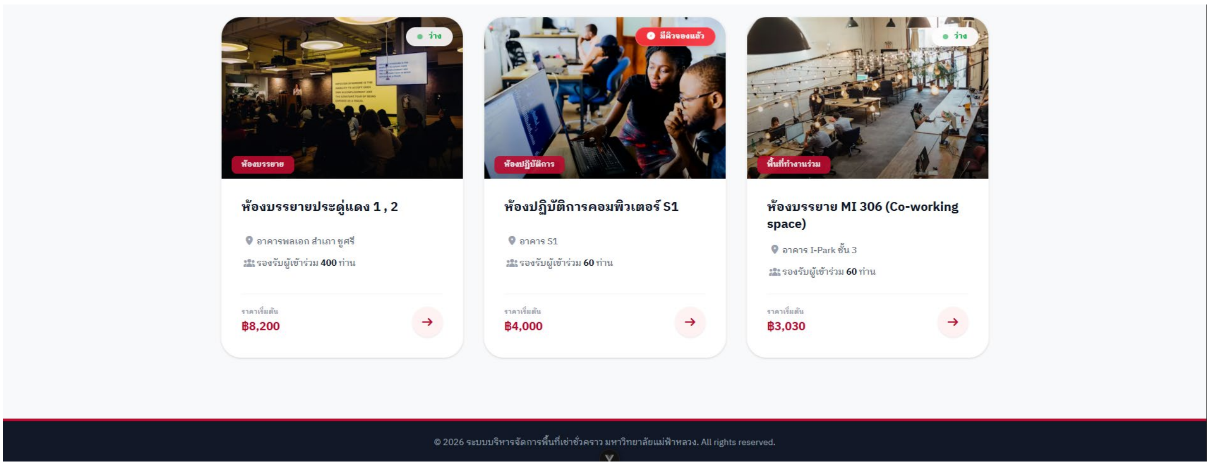

Figure 3.13 User interface for all-room page

The room details page could be demonstrated by a user interface in **Figure 3.14 - Figure 3.15**

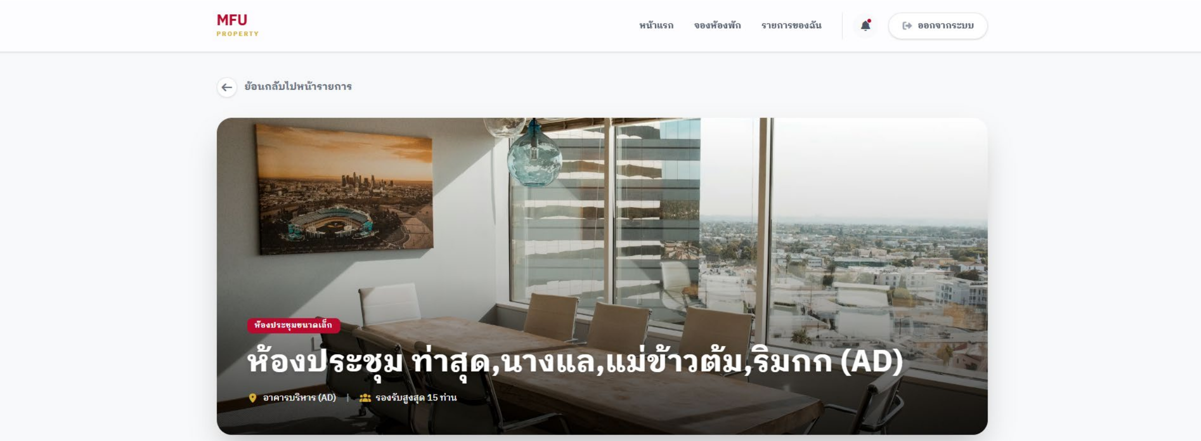

Figure 3.14 User interface for room details page

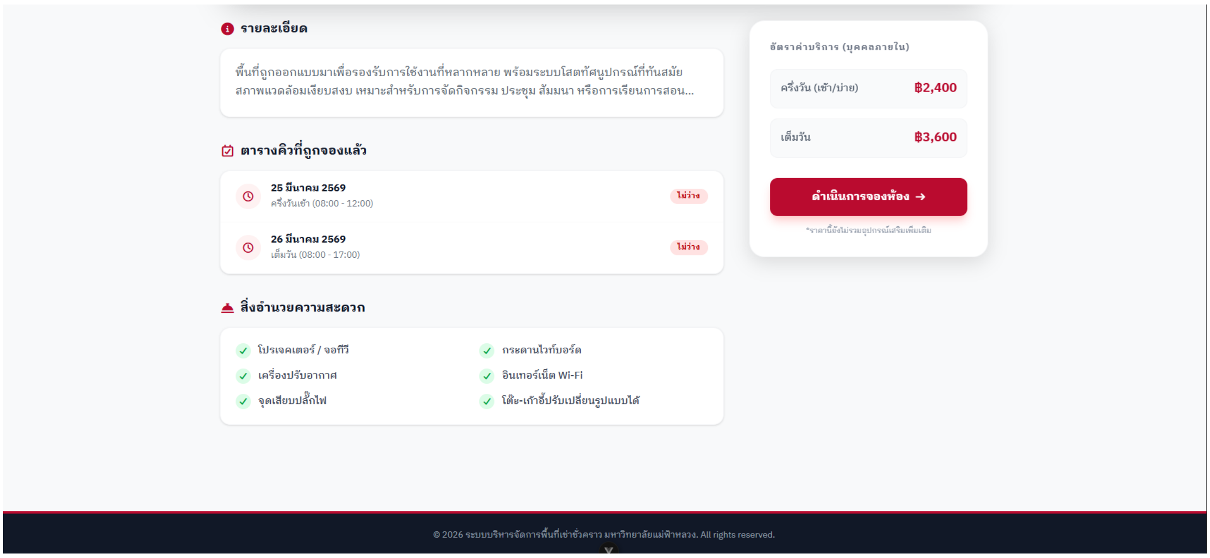

Figure 3.15 User interface for room details page

The booking room page could be demonstrated by a user interface in **Figure 3.16 - Figure 3.17**

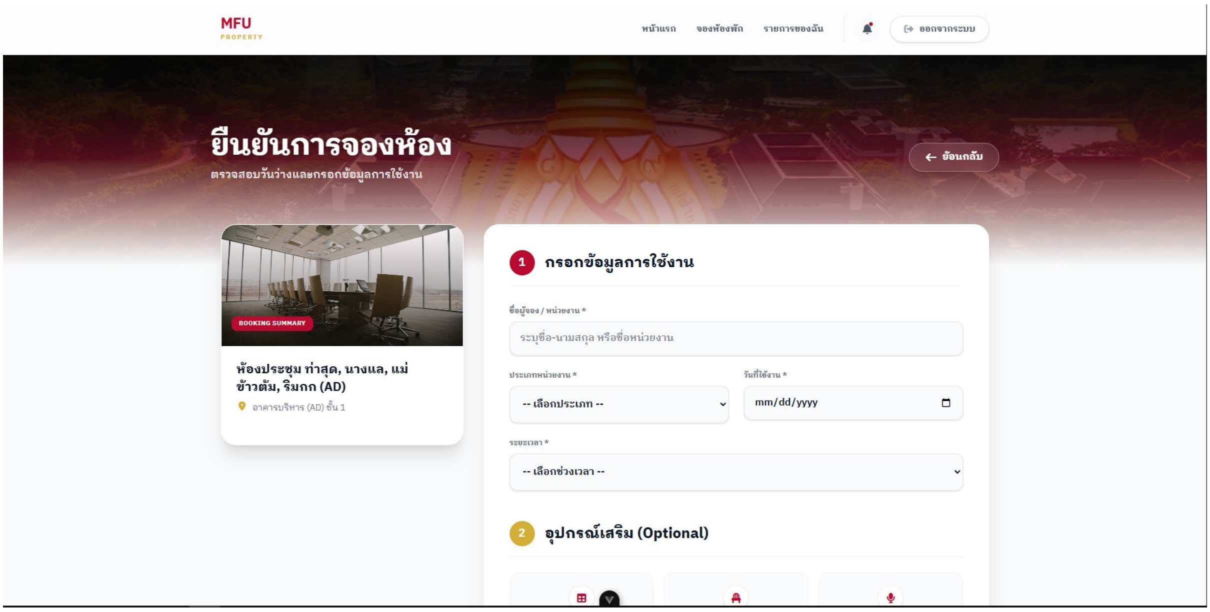

Figure 3.16 User interface for booking room page

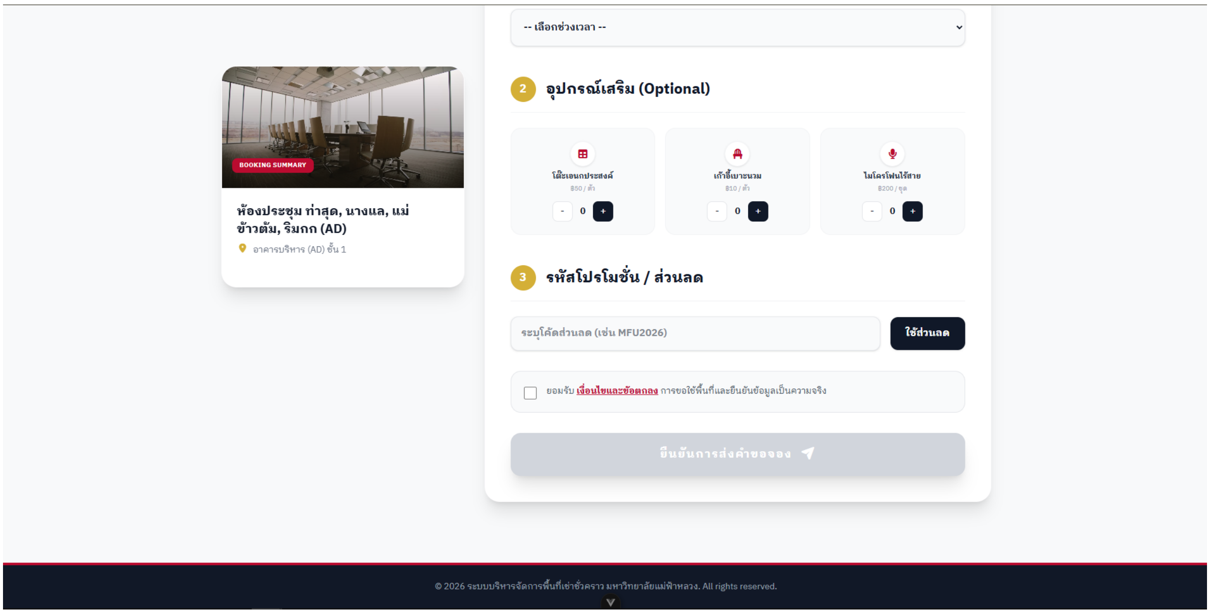

Figure 3.17 User interface for booking room page

The user dashboard page could be demonstrated by a user interface in **Figure 3.18 - Figure 3.19**

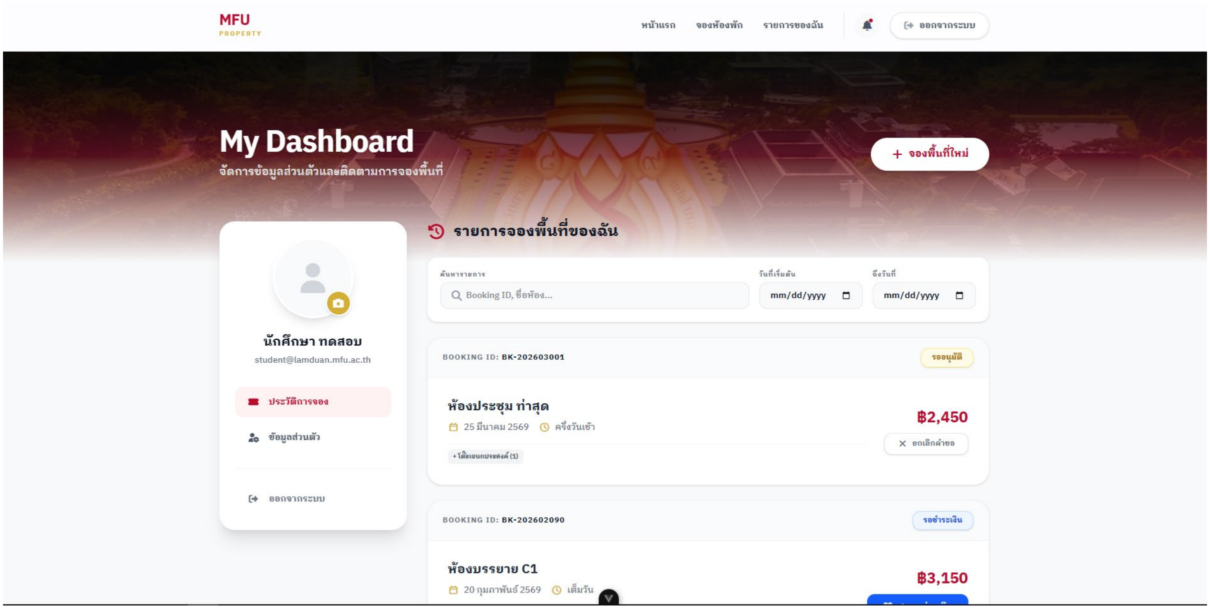

Figure 3.18 User interface for user dashboard page

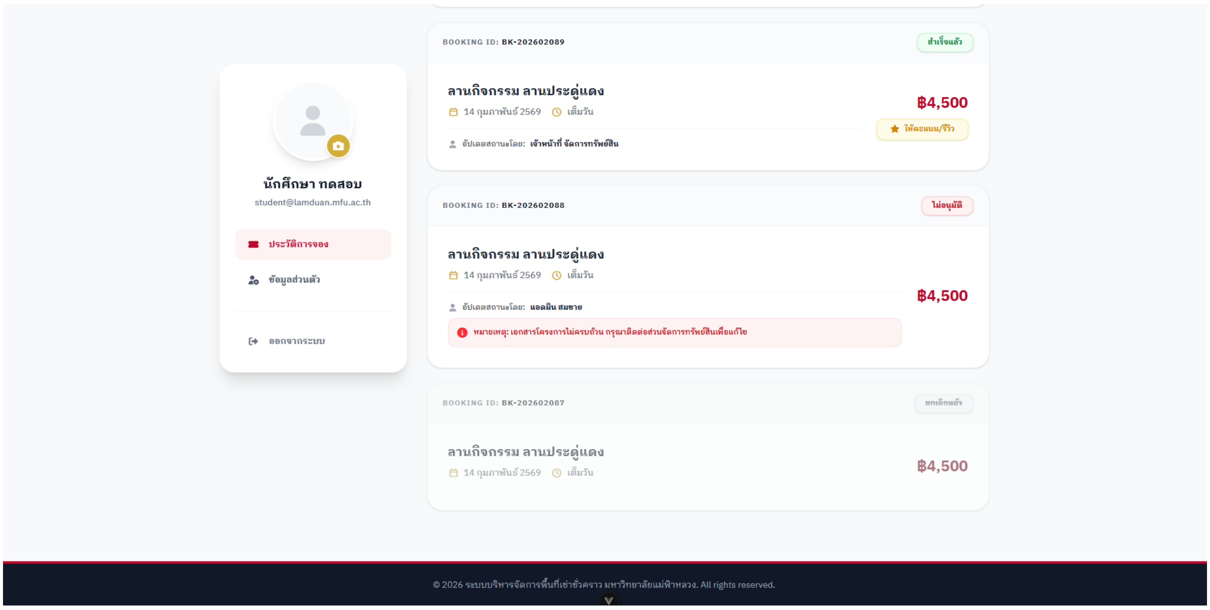

Figure 3.19 User interface for user dashboard page

The privacy information page could be demonstrated by a user interface in **Figure 3.20**

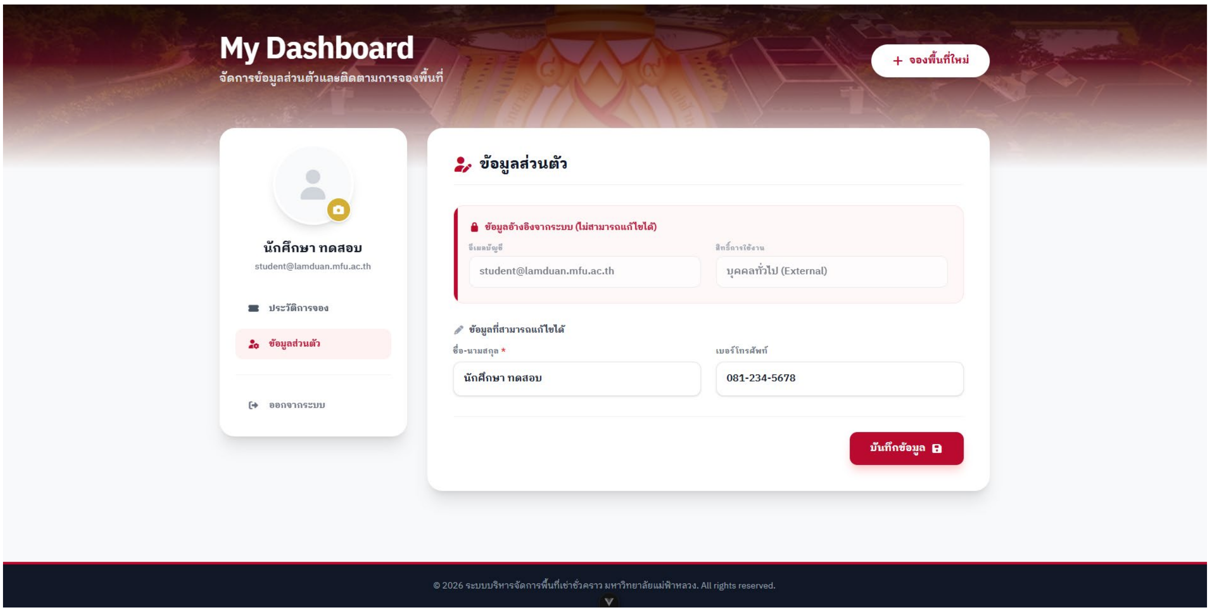

Figure 3.20 User interface for privacy information page

The admin dashboard page could be demonstrated by a user interface in **Figure 3.21 - Figure 3.22**

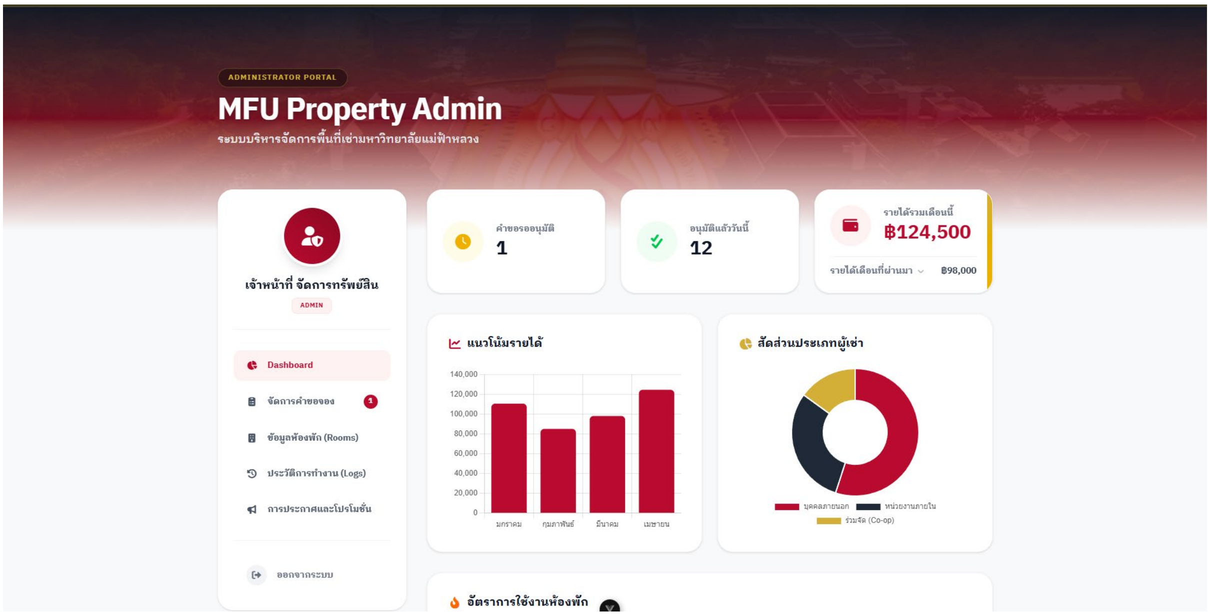

Figure 3.21 User interface for admin dashboard page

Figure 3.22 User interface for admin dashboard page

The admin request page could be demonstrated by a user interface in **Figure 3.23 - Figure 3.24**

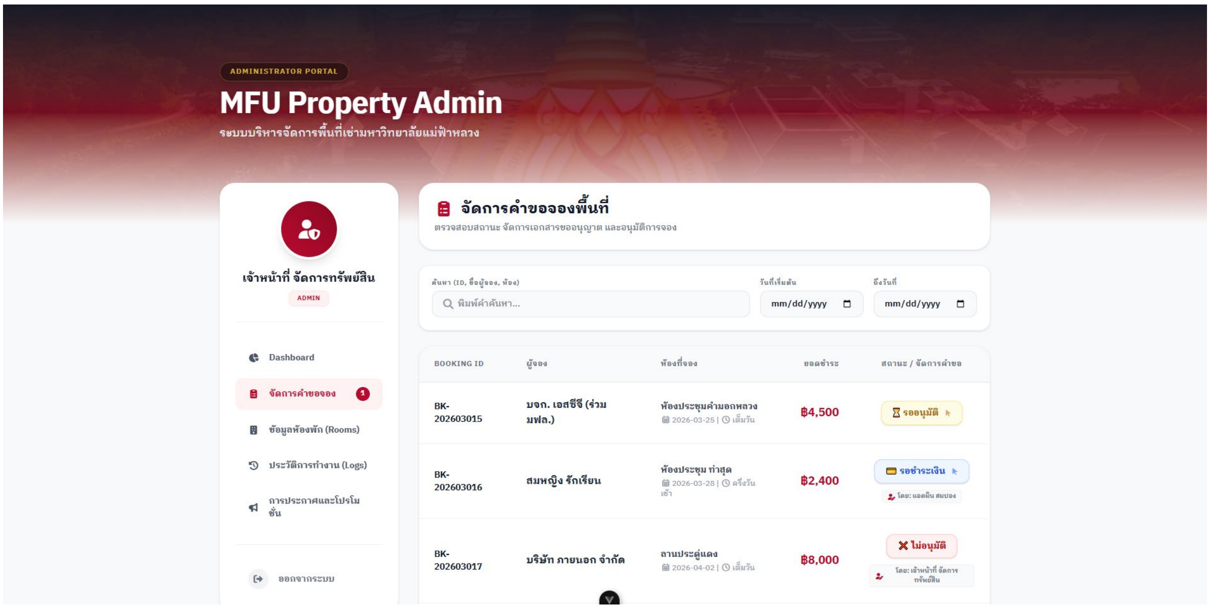

Figure 3.23 User interface for admin request page

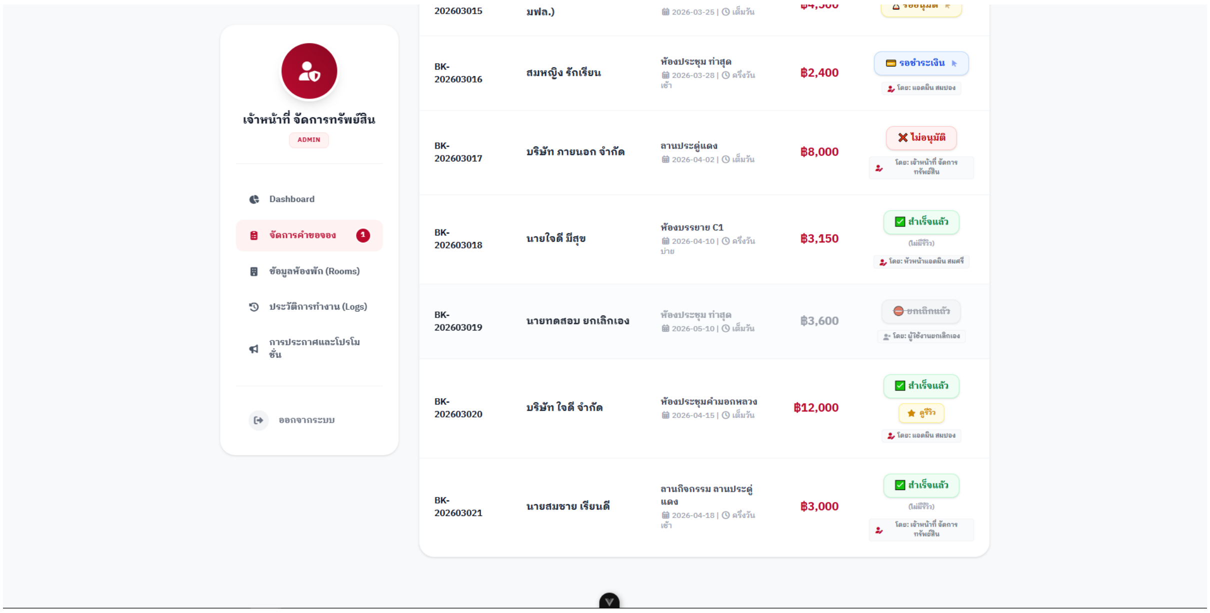

Figure 3.24 User interface for admin request page

The admin edit room page could be demonstrated by a user interface in **Figure 3.25**

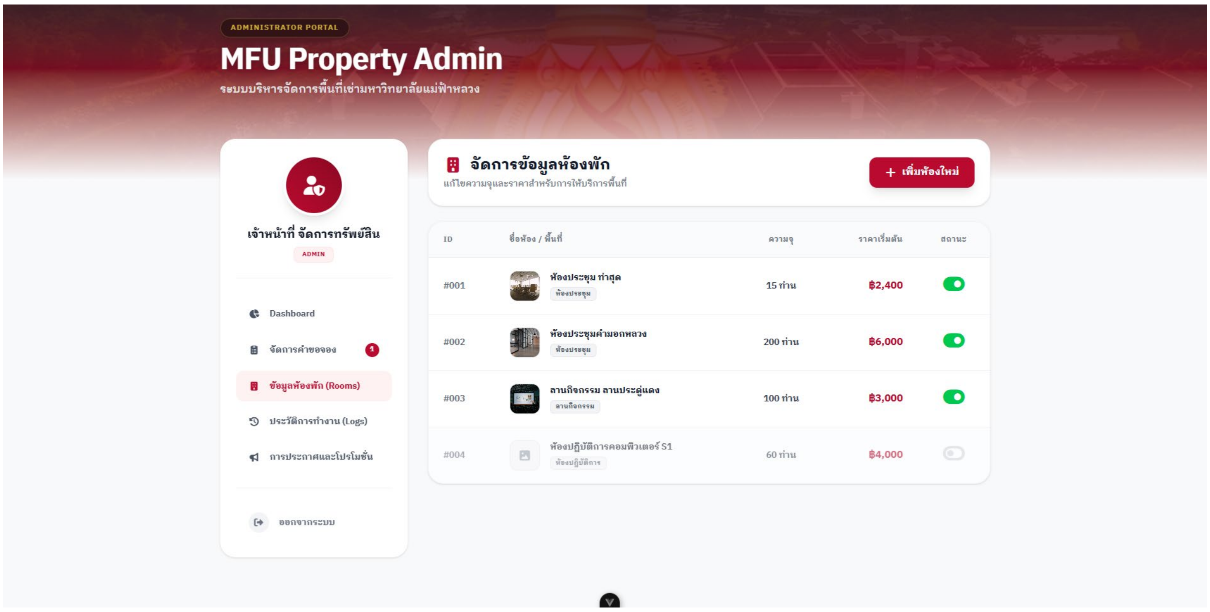

Figure 3.25 User interface for admin edit room page

The admin history activity could be demonstrated by a user interface in **Figure 3.26**

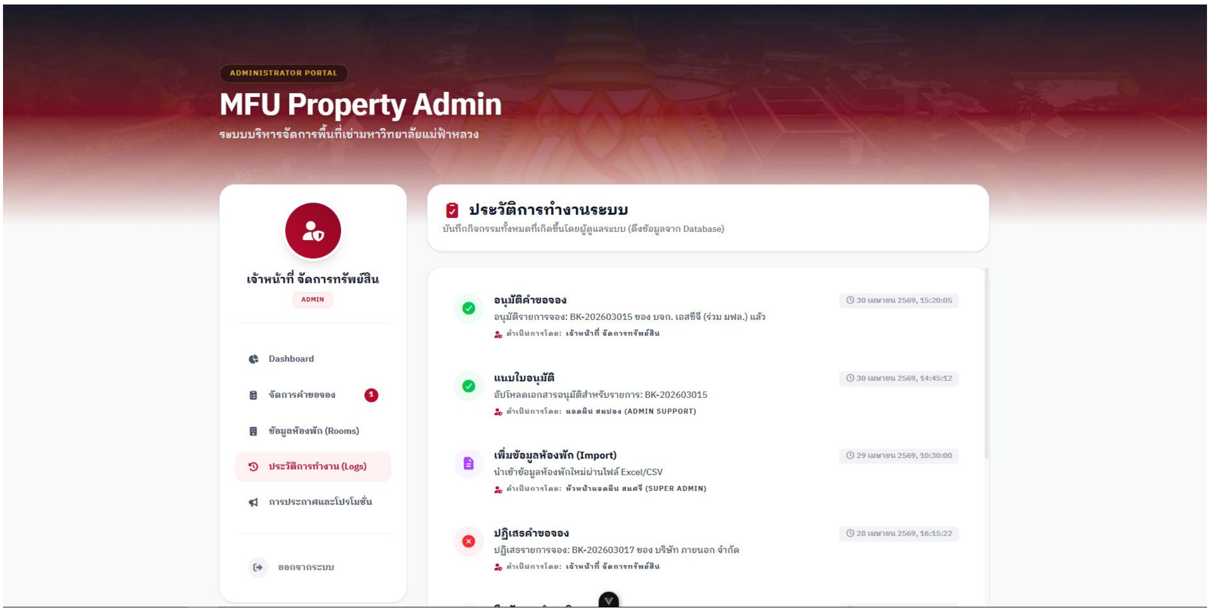

Figure 3.26 User interface for admin history activity page

The admin broadcast and promotions page could be demonstrated by a user interface in **Figure 3.27 - Figure 3.28**

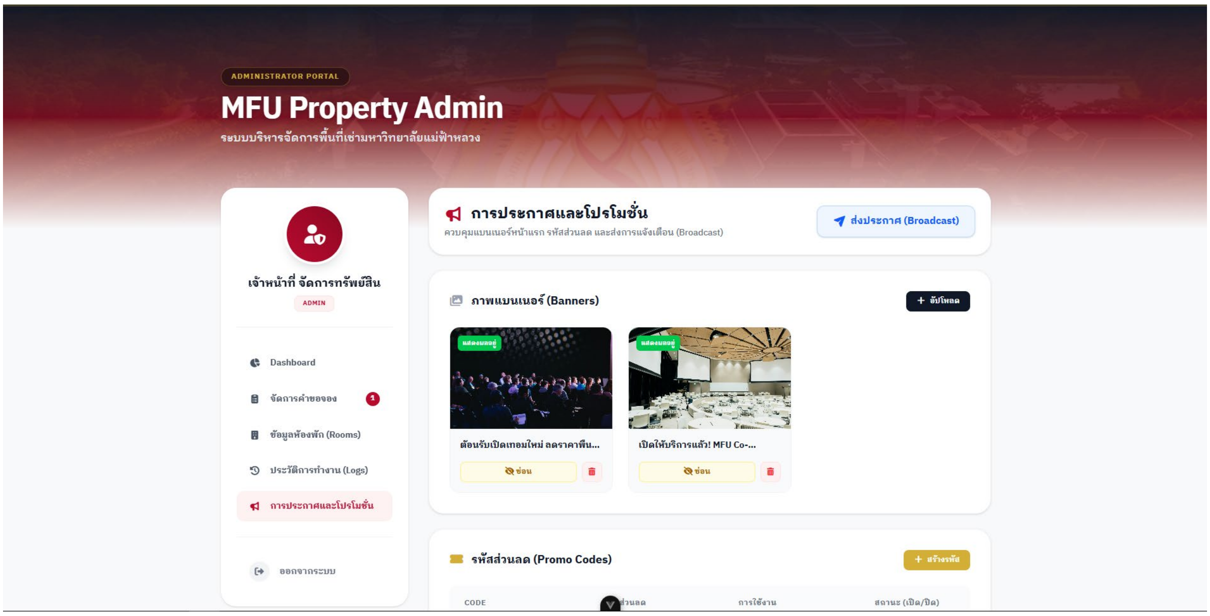

Figure 3.27 User interface for admin broadcast and promotions page

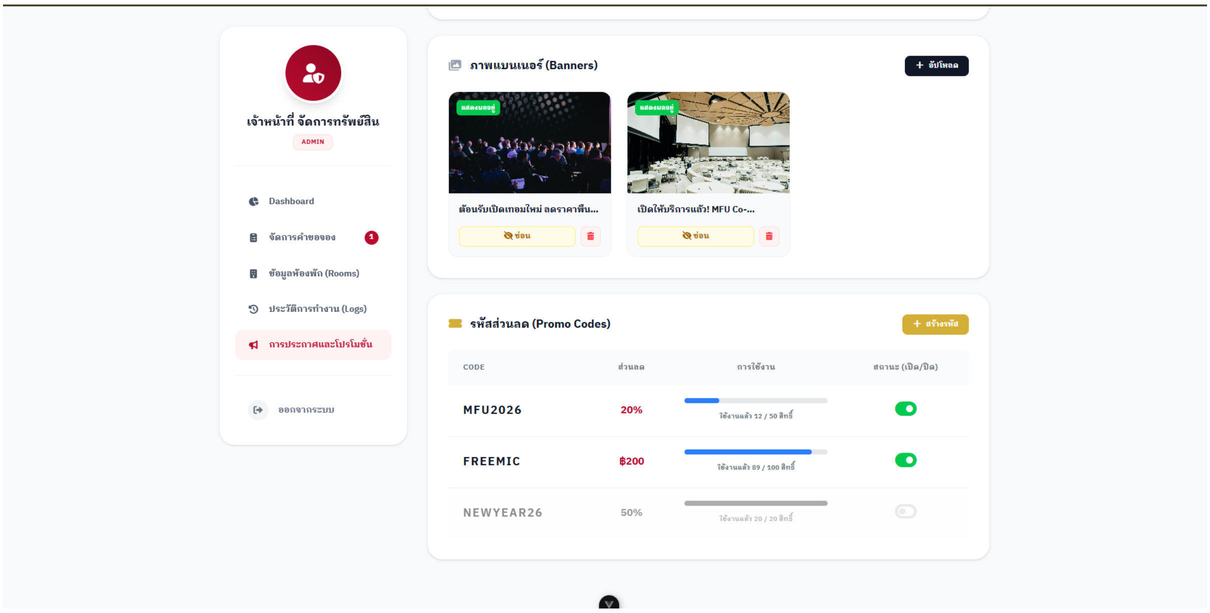

Figure 3.28 User interface for admin broadcast and promotions page

3.5 ER diagram

The database could be demonstrated by an er diagram in **Figure 3.29**

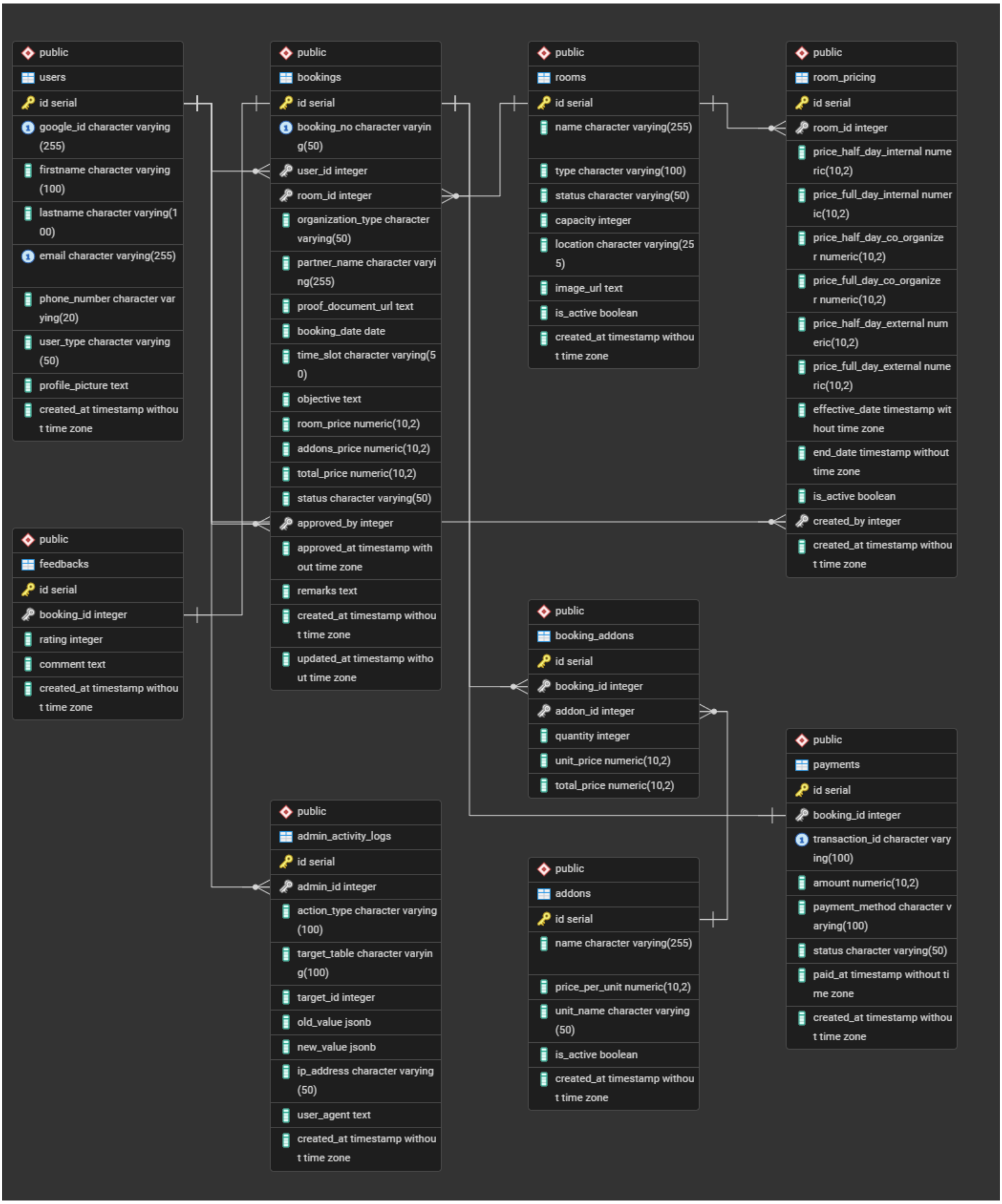

Figure 3.29 ER diagram for database

3.6 Data Dictionary

The database could be demonstrated by a data dictionary in **Figure 3.30 – Figure 3.38**

Field Name	Data Type	Field Length	Constraint	Description	Example
id	serial	-	PK	Unique identifier for each feedback	1
booking_id	integer	-	FK	Reference to booking	1
rating	integer	-	-	Rating (1–5)	5
comment	text	-	-	Feedback or review provided by the use	Great service
created_at	timestamp without time zone	-	-	Record creation time	27/04/2026 14:30:31

Figure 3.30 Data Dictionary for feedback table

Field Name	Data Type	Field Length	Constraint	Description	Example
id	serial	-	PK	Booking Add-on ID	1
booking_id	integer	-	FK	Reference to Bookings table	1
addon_id	integer	-	FK	Reference to Addons table	1
quantity	integer	-	-	Quantity of add-on	2
unit_price	numeric	10,2	-	Price per unit at time of booking	100.00
total_price	numeric	10,2	-	Total price	200.00

Figure 3.31 Data Dictionary for booking addons table

Field Name	Data Type	Field Length	Constraint	Description	Example
id	serial	-	PK	Payment ID	1
booking_id	integer	-	FK	Reference to Bookings table	1
transaction_id	character varying	100	Unique	Transaction ID	TXN123456
amount	numeric	10,2	-	Payment amount	1200.00
payment_method	character varying	100	Unique	Payment method	promptpay
status	character varying	50	-	Payment status	paid
paid_at	timestamp without time zone	-	-	Payment timestamp	03/01/2026 09:00:00
created_at	timestamp without time zone	-	-	Record creation time	03/01/2026 12:00:00

Figure 3.32 Data Dictionary for payments table

Field Name	Data Type	Field Length	Constraint	Description	Example
id	serial	-	PK	Add-on ID	1
name	character varying	255	-	Name of the add-on	Projector
price_per_unit	numeric	10,2	-	Price per unit	100.00
unit_name	character varying	50	-	Unit of measurement	piece
is_active	boolean	-	-	Active status	TRUE
created_at	timestamp without time zone	-	-	Record creation time	03/01/2026 10:00:00

Figure 3.33 Data Dictionary for addons table

Field Name	Data Type	Field Length	Constraint	Description	Example
id	serial	-	PK	Room ID	106
name	character varying	255	-	Room name	Building C3, 2nd floor
type	character varying	100	-	Type of room	Meeting Room
status	character varying	50	-	Room status	available, unavailable
capacity	integer	-	-	People capacity	200
location	character varying	255	-	Location	C3 2nd floor
image_url	text	-	-	Image URL	room1.jpg
is_active	boolean	-	-	Active status	TRUE
created_at	timestamp without time zone	-	-	Creation date	03/01/2026 10:00:00

Figure 3.34 Data Dictionary for rooms table

Field Name	Data Type	Field Length	Constraint	Description	Example
id	serial	-	PK	Pricing ID	1
room_id	integer	-	FK	Reference to room	1
price_half_day_internal	numeric	10,2	-	Half-day price (internal users)	500.00
price_full_day_internal	numeric	10,2	-	Full-day price (internal users)	500.00
price_half_day_co_organizer	numeric	10,2	-	Half-day price (co-organizer)	700.00
price_full_day_co_organizer	numeric	10,2	-	Full-day price (co-organizer)	1200.00
price_half_day_external	numeric	10,2	-	Half-day price (external users)	1000.00
price_full_day_external	numeric	10,2	-	Full-day price (external users)	1800.00
effective_date	timestamp without time zone	-	-	Effective start date	03/01/2026 10:00:00
end_date	timestamp without time zone	-	-	Effective start date	03/01/2026 10:00:00
is_active	boolean	-	-	Active status	TRUE
created_by	integer	-	FK	Created by (user ID)	1
created_at	timestamp without time zone	-	-	Record creation timestamp	03/01/2026 10:00:00

Figure 3.35 Data Dictionary for room pricing table

Field Name	Data Type	Field Length	Constraint	Description	Example
id	serial	-	PK	Booking ID	1001
booking_no	character varying	50	Unique	Booking number	BK20260001
user_id	integer	-	FK	Reference to Users table	1002
room_id	integer	-	FK	Reference to Rooms table	12
organization_type	character varying	50	-	Organization type	internal / external / co-organizer
partner_name	character varying	255	-	Partner organization name	ABC Co., Ltd
proof_document_url	text	-	-	Proof document URL	doc.pdf
booking_date	date	-	-	Booking date	1/5/2026
time_slot	character varying	50	-	Time slot	morning / afternoon / full-day
objective	text	-	-	Booking purpose	Team meeting
room_price	numeric	10,2	-	Room price	1000.00
addons_price	numeric	10,2	-	Add-ons total price	1200.00
total_price	numeric	10,2	-	Total price	2000.00
status	character varying	50	-	Booking status	approved
approved_by	integer	-	FK	Approved by (user ID)	2
approved_at	timestamp without time zone	-	-	Approval timestamp	01/01/2026 10:00:00
remarks	text	-	-	Additional remarks	-
created_at	timestamp without time zone	-	-	Record creation timestamp	03/01/2026 10:00:00
updated_at	timestamp without time zone	-	-	Record update timestamp	04/01/2026 10:00:00

Figure 3.36 Data Dictionary for bookings table

Field Name	Data Type	Field Length	Constraint	Description	Example
id	serial	-	PK	Unique identifier for activity log	1
admin_id	integer	-	FK	Reference to admin user	2
action_type	character varying	100	-	Type of action performed	update
target_table	character varying	100	-	Target table affected	bookings
target_id	integer	-	-	ID of affected record	1
old_value	jsonb	-	-	Previous data (before change)	{...}
new_value	jsonb	-	-	Updated data (after change)	{...}
ip_address	character varying	50	-	IP address of user	127.0.0.1
user_agent	text	-	-	Browser or device information	Chrome
created_at	timestamp without time zone	-	-	Record creation timestamp	-

Figure 3.37 Data Dictionary for admin activity log table

Field Name	Data Type	Field Length	Constraint	Description	Example
id	serial	-	PK	Unique identifier for each user	1001
google_id	character varying	255	-	Google account ID	123abc
firstname	varchar varying	100	-	User Firstname	Somchai
lastname	varchar varying	100	-	User Lastname	Jaidee
email	character varying	255	Unique	User email address	Somchai007@gmail.com
phone_number	character varying	20	-	User phone number	812345678
user_type	character varying	50	-	Type of user	admin, user
profile_picture	text	-	-	Image URL	img.jpg
created_at	timestamp without time zone	-	-	Timestamp when the user account was created	03/01/2026 10:00:00

Figure 3.38 Data Dictionary for user table

3.7 Third-Party Service Integrations

The system integrates two major third-party services to enhance security and streamline administrative workflows:

- 3.7.1 Google OAuth 2.0 Authentication The system utilizes Google OAuth 2.0 for Single Sign-On (SSO).
  - Integration: The Node.js backend securely retrieves user profiles via Google's authorization server.
  - Automated Role Assignment: The system assigns roles based on email domains. Emails ending in @mfu.ac.th or @lamduan.mfu.ac.th are registered as "Internal Users," while public domains (e.g., @gmail.com) are "External Users."
- 3.7.2 Payment Gateway Integration To automate payment verification, the system integrates a payment gateway to generate dynamic PromptPay QR codes.
  - Gateway Selection: Opn Payments was selected over 2C2P due to its developer-friendly API, comprehensive Node.js libraries, and an accessible Sandbox environment.
  - Transaction Fee: Opn Payments charges a 1.65% fee (exclusive of VAT) per successful transaction, which must be accounted for in the system's revenue management.
  - Payment Workflow:
    1. The Node.js backend requests a dynamic PromptPay QR Code from the Opn Payments API for the total price.
    2. The Vue.js frontend displays the generated QR code.
    3. After the user pays via mobile banking, Opn Payments sends a secure Webhook to the server.
    4. The system validates the Webhook and automatically updates the booking status to "Paid."

CHAPTER 4
CONCLUSION

4.1 Completed Work (January - May 2026)
During the first phase of the project (Senior Project 1), the team successfully established the foundational structure of the Temporary Rental Space Management System. The completed tasks include:

- Requirement Analysis: Gathered and analyzed user requirements based on the existing workflows of the Property Management Division at Mae Fah Luang University.
- System Design: Designed the overall system architecture, relational database schema (PostgreSQL), and user interfaces (UI).
- Prototyping: Developed the initial functional prototype of the web application to demonstrate the core user flows and centralized dashboard concepts.
- Documentation: Prepared the formal Project Proposal and completed the Senior Project 1 examination.

4.2 Future Work Plan (August - December 2026)
For the next phase in Senior Project 2, the team will focus on the full implementation and deployment of the system. The planned tasks include:

- System Development: Fully implement the full-stack web application, seamlessly integrating the Vue.js frontend with the Node.js backend logic.
- Third-Party Integrations: Implement Google OAuth for secure single sign-on (SSO) and integrate the Opn Payments gateway to enable PromptPay QR transactions.
- System Testing and Evaluation: Conduct rigorous software testing to resolve bugs, validate role-based access controls, and ensure the system accurately prevents scheduling conflicts.
- Final Defense: Prepare the official Senior Project 2 documentation and present the finalized system to the examination committee.